MILITARY SPECIFICATION

COMPOSITION, TUNGSTEN, DELAY

1. SCOPE

1.1 Scope.-This specification covers one type of tungsten delay composition for use in the loading of time fuzes (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-T-48140 - Tungsten Powder (For Pyrotechnics)
MIL-P-217 - Potassium Perchlorate
MIL-B-550 - Barium Chromate
MIL-V-50433 - Vinyl Alcohol-Acetate Resin Solution (VAAR) (For Use in Ammunition)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1234- Pyrotechnics: Sampling, Inspection and Testing

DRAWINGS

ARMY

FSC: 1376

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9219355 - Fuze, Time, M84A1
9232791 - Graduated Time Train Ring Assembly
9232792 - Upper Time Train Ring Assembly

PUBLICATIONS

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

Title 49 - Transportation Part 0 - 190

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite 49 CFR 0-190 (Latest revisions).

3. REQUIREMENTS

3.1 Material.-The constituent materials used in the manufacture of the powders purchased under this specification shall comply with following requirements:

3.1.1 Barium Chromate.-The Barium Chromate shall comply with Specification MIL-B-550, Grade C.

3.1.2 Potassium Perchlorate.-The Potassium Perchlorate shall comply with Specification MIL-P-217, Grade A, Class IV.

3.1.3 Tungsten.-The Tungsten shall comply with Specification MIL-T-48140, Type II.

3.1.4 Vinyl Acetate Alcohol Resin (VAAR).-The VAAR shall comply with Specification MIL-V-50433.

3.2 Composition.-The tungsten composition shall be as specified in Table I when tested as specified in 4.4.5. (see 6.6).

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TABLE I

Constituents	Percent
Tungsten (Nominal	58
Barium Chromate (Nominal)	32
Potassium Perchlorate	10 ± 0.5
VAAR*	*12. (Advisory)

*Parts by weight, solids on added basis see 6.6.

3.3 Volatiles.-The volatiles content of the composition shall be no more than 0.1 percent when tested as specified in 4.4.1.

3.4 Granulation.-The granulation of the tungsten delay composition shall be as specified in Table II when determined as specified in 4.4.2.

TABLE II

USS Sieve No.	Percent Retained
20	3.00 Max.
80	91.00 Min.

3.5 Gritty or foreign particles.-The composition shall be free of gritty particles when determined as specified in 4.4.3.

3.6 Static burning time.-The static burning time shall be as specified in Table III when tested as specified in 4.4.4. (see 6.5).

TABLE III

Fuze	Average Burning Time (Seconds)	Range About the Average (Seconds)
M84A1	48 ± 8	± 3

3.7 First article testing.-This specification makes provisions for first article testing. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such materials is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 1 pound of delay composition in accordance

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with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted wnenever there is a lapse in production for a period of excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect production uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4 of this spefication.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of one or more batches, of delay composition produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of delay composition that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.3.2 Testing

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4.3.2.1 Static Burning Time, Major Code 04001.-Ten samples weighing 100 grams each shall be randomly selected from individual containers. The individual samples shall be packed separately in suitable containers and shipped to the testing laboratory specified in the contract (see 6.5). When the lot is comprised of less than ten containers, all containers shall be samples. If the average burning time or an individual times about the average fails to comply with the requirement of 3.6, the lot shall be rejected.

4.3.2.2 Sampling for chemical test.-Five samples weighing 100 grams each shall be randomly selected from individual containers. The individual samples shall be blended to form two composite samples. The two composited samples shall be tested in accordance with paragraph 4.3, through 3.5, the lot shall be rejected. The following defects shall be used for the test specified.

Tests	Defects	Code No.
Volatiles	Major	01001
Granulation	Major	02001
Gritty or foreign particles	Minor	03001
Potassium Perchlorate	Major	05001

4.4 Test procedures and methods

4.4.1 Volatiles.-The volatiles shall be determined in accordance with method 102.2 of Standard MIL-STD-1234.

4.4.2 Granulation.-The granulation shall be determined in accordance with method 201.1 of Standard MIL-STD-1235. Except a 100 gram sample shall be used and the shaker shall be operated for one minute without the striker.

4.4.3 Gritty or foreign particles.-The composition shall be examined for foreign particles by visual examination.

4.4.4 Determination of Static Burning Time.-Spread the powder in trays to a depth of approximately 1/16 inch and condition in an oven or drying room for two hours minimum at a temperature of 90°F minimum. The time train rings of twenty M84A1 Time Fuzes shall be loaded with powder (dwgs. 9232791 and 9232792) and assembled in accordance with drawing 9232784. The graduated ring shall be set 50 seconds and the fuze shall be tested statically at ambient temperature of 75 ± 5°F. An approved static burning time fuze tester, which can be read directly to at least the nearest 0.05 second, shall be used.

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4.4.5 Potassium Perchlorate.-Transfer an accurately weighed portion of approximately one gram of the sample to a weighed Gooch crucible. Leach with ten 10 ml. portions of hot water and place the crucible, including sample, in a drying oven at 100 degrees to 105 degrees C. for two hours. Cool in a desiccator and weigh the crucible with the residue. The percent potassium perchlorate shall be calculated as follows:

$$\text{Percentage Potassium Perchlorate} = \frac{100.0\ (A)}{W}$$

Where:

A = loss in weight of the sample
B = weight of sample on a moisture-free basis

5. PREPARATION FOR DELIVERY

5.1 Packing-Level C.-Delay composition shall be packed in metal containers, containing no more than 100 pounds gross weight, unless otherwise specified by the procuring agency, in such a manner as to comply with the applicable regulations of the Interstate Commerce Commission (see Code of Federal Regulations, 49-CFR 0-190 and as to insure acceptance by common or other carrier for safe transporation at the lowest rate to the point of delivery.

5.2 Marking.-All containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Intended Use.-The delay composition is intended for use in the M84A1 Fuze.

6.4 Test results.-A copy of the test results should be forwarded to Commanding Officer, ATTN: SMUPA-QA-A-P, Picatinny Arsenal, Dover, New Jersey 07801.

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6.5 Burning time test.-The contracting officer should request the loading plant which has the fuze production contract to perform the burning time test.

6.6 Process Procedure

6.6.1 The delay composition shall be manufactured using conventional wet blending techniques and employ ethyl alcohol, Grade 3, or Grade 5, Specification MIL-E-463, as a blending medium.

6.6.2 Pilot mixture (Advisory). The burning time of delay compositions manufactured from different lots or suppliers of ingredients may vary. A pilot mixture technique is therefore advisory to define the composition-burning time curve for new lots or suppliers of ingredients.

6.6.3 Vinyl Alcohol Acetate Testing (VAAR) solids determination.-The concentration of the solids in the VAAR solution may change upon storage due to the evaporation of solvent. The VAAR solids content in the delay composition meeting the requirement specified in 3.2 can only be obtained if the solution to solidweight ratio for the VAAR solution is known. A sample of the VAAR solution should be periodically tested per MIL-V-50433 to determine this ratio.

6.7 Static Burning Time (see 4.4.3).-The 2 hours conditioning requirement prior to static testing of the tungsten delay powder may be deleted if the time interval between process drying and test is less than 24 hours.

6.8 Supersession Data.-This specification includes the requirements of Purchase Description PA-PD-3091 dated 29 March 1968.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1376-A003

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Department of the Army
Headquarters, Picatinny Arsenal
Dover, New Jersey 07801

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